



Spatial and temporal variability of forest fires in the Republic of Korea over 1991–2020

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Abstract

Forest fires have increased over the last several decades in many regions. Quantifying the general patterns of frequency, areal extent, and seasonality is crucial for understanding fire dynamics. This study aimed to investigate whether the spatial and temporal trends in forest fires have changed across South Korea. The Mann–Kendall test and Sen’s slope estimation were used to analyze the temporal trends in forest fire statistics from 1991 to 2020. The spatial dispersion of fire activity was detected using a standard deviation ellipse and hotspot analysis. An average of 451 fires have occurred annually over the last 30 years, with a yearly increase of 5.82 fires. The burned area in April and May accounted for 80.7% of the annual burned area. The length of the fire season in 2006–2020 was 25 days longer than that in 1991–2005. The risk of large fires is increasing and becoming more concentrated in the northeastern region, such as the Gwangwon and Gyeongsangbuk Provinces of South Korea. Both climate change and forest recovery have led to South Korea becoming more prone to fires. However, forest fires are not burning more intensely nor charring more areas than they did previously. This is probably due to the implementation of surveillance and initial attack systems. Targeted forest fire suppression policies can help to effectively reduce the risk of forest fires in South Korea.

Keywords Forest fire · Fire occurrence · Hotspot analysis · Standard deviation ellipse · Fire season · Burned area · Spatial variability · Temporal variability

1 Introduction

Forest fires are among the most destructive disasters and pose a serious, growing threat to forest ecosystems worldwide. Fires shape the structure and function of the forest landscape by completely or partially removing aboveground vegetation. The frequency and magnitude of forest fires have increased over the last several decades in many regions (Prichard et al. 2017; Jones et al. 2020), but with high interannual variability. Globally, forest fires occur at over 200,000 locations annually, with burnt areas covering approximately 3.5–4.5 million km² (Jones et al. 2020).

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The Republic of Korea (hereafter, South Korea) has historically faced frequent forest fires. Forests cover approximately 63.2% of the total land area of South Korea, ranking fourth-highest among OECD countries. Following the colonial period and the Korean War (1950–1953), approximately one-third of forests were removed or denuded, and consequently, in the 1960s, the forest volume dropped to approximately 6 m³/ha. However, owing to the National Forest Rehabilitation Plans that started in 1973, the growing stock increased to 168.67 m³/ha by 2021 (Korea Forest Service 2022). Successful reforestation can lead to increased surface fuel and thicker litter layers in forests, which contribute to an elevated risk of forest fires, particularly in highly ignitable pine forests. For instance, according to the National Fire Statistics of South Korea, four of the largest fire years on record have occurred since 2000 (Li et al. 2021; Jeon and Chae 2017).

The spatial and temporal patterns of fires have recently been noted on a global scale (Shi and Touge 2023) and in many countries, such as China (Zhang et al. 2016), Portugal (Carmo et al. 2022), Canada (Stocks et al. 2002; Hanes et al. 2019), Spain (Díaz-Delgado et al. 2004), and the United States (Miller and Safford 2012). Studies have also reported increases in burned areas and the lengthening of the fire-prone season over the last few decades (Pimont et al. 2023; Jones et al. 2022). This trend is expected to increase further over the next century due to climate change.

Climate influence on fire ignition, fuel moisture and the environment in which the fire burns. Previous studies have addressed increasing trends in fire frequency and severity worldwide in recent decades and have also predicted increases in fire risk under future climate change (Canadell et al. 2021; Carmo et al. 2022; Dastour et al. 2024). Canadell et al. (2021) found that dry and hot weather has led to more frequent fire activity in Australia by increasing fuel dryness, but fuel management has had no effect on the long-term increasing trend in forest area burned. Stocks et al. (1998) also conducted a monthly analysis, which showed an earlier start to the fire season and a significant increase in the area of high to extreme fire danger in both Canada and Russia. In addition, large increases in forest fire activity have been documented globally for American forests (Chen et al. 2011; Feltman et al. 2012), Mediterranean forests (Bountzouklis et al. 2022; Carmo et al. 2022; Miller and Safford 2012), boreal forests (Kelly et al. 2013), subtropical pine forests (Mitchell et al. 2014), and tropical rainforests and savannas (Brando et al. 2014). While longer-term (decadal or longer) climate variability can have a major impact on fuel types and loads, the response in terms of fire frequency and severity is highly variable as this is a spatially dependent relationship.

Several approaches are available to detect long-term changes in fire features from historical fire records. However, the Sen slope estimator (Sen 1968) and the Mann–Kendall trend test (Kendall 1975) are the most commonly used to quantify temporal trends in fire activity due to several advantages such as dealing with missing data, making few assumptions and being independent of the data distribution (Barthel et al. 2015; Esterby 1996). Hotspot mapping has become increasingly popular in forest fire management studies, where it can help to assess the spatial distribution of fire incidents and identify fire risk at a large scale (Dastour et al. 2024).

Hotspot analysis is a mapping technique for identifying the spatial autocorrelation of events within a limited geographical area (Barthel et al. 2015). There are many methods of hotspot analysis, each with its own advantages and disadvantages. Common methods include Moran's I, Kernel Density Estimation (KDE) and Getis-Ord Gi*. Moran's I statistic

can detect the presence of clustering of similar values in a geospatial dataset (Moran 1950), but cannot quantify whether the clustering is of high or low values (Mtshawu et al. 2023). In contrast, both KDE (Silverman 1986) and Getis-Ord G_i^* statistic (Getis and Ord 1992; Ord and Getis 1995) can identify significant spatial clusters of high (hot spot) and low (cold spot) values. Ganteaume and Guerra (2018) used KDE to obtain hotspot maps of fire ignition with different fire causes in France. Based on the identified hotspots, they found that undetermined arson and negligence during agricultural work were the most damaging and frequent causes in the study area. Feltman et al. (2012) used another hotspot analysis method, known as Getis-Ord G_i^* , to identify socio-economic variables that contribute to fire occurrence. Site-specific characteristics have a strong, but complex, influence on fire activity, which varies across time and space. Additionally, changes in the climate between years can differentially affect fire patterns. Quantifying the general patterns of frequency, areal extent, and seasonality of fires is crucial for understanding the past and future trends of fire dynamics. Such analysis becomes necessary during the planning or implementation of appropriate mitigation, preparedness, and response activities for fire (Short 2014). However, gaining a systematic understanding of the variability in fire activity remains challenging in many countries, including South Korea, because of the complex interactions among climate, meteorology, vegetation, humans, and fires (Jones et al. 2022).

The general objective of this study was to investigate whether the spatial and temporal trends in forest fire characteristics have changed across South Korea. More specifically, the long-term trends in fire occurrence and burned area were analyzed. Furthermore, spatial and temporal analyses of the number of fires and burned area were conducted for the whole period of 1991–2020 and the two sub-periods of 1991–2005 and 2006–2020 separately. The results are discussed in the context of forest fire suppression strategies and climate change, and whether these may be attributed to fire regime changes.

2 Materials and methods

2.1 Data set

South Korea is a country of East Asia, occupying the southern part of the Korean peninsula. The country is characterized by a temperate monsoon climate with four distinct seasons. The hot and wet summer season occurs from June to August, whereas the winter is dry and cold and occurs from December to February. The spring (March–May) and fall (September–November) seasons are transition periods between the summer and winter. The annual precipitation over the country is 1306 mm, and two-thirds of the precipitation occurs during the summer season between June and August. The average mean temperature varies from 23 to 27 °C in summer and −6 to −7 °C in winter. The atmospheric humidity is highest in July, reaching 80–90% nationwide. In contrast, the lowest mean monthly humidity is 30–50% in January and April (Korea Meteorological Administration 2023).

Much of the forest in South Korea were severely degraded and depleted by excessive logging during the first half of the 20th century. Since 1973, the Government has undertaken a multi-year forest restoration project, which has resulted in the successful recovery of the forest. Thanks to the national reforestation effort, the average forest stock has increased from 11.31 m³/ha in 1973 and 176 m³/ha in 2023. The country's forest area spans

6.3 million ha in 2023, which accounts for 63.1% of the country's total land area. The forest features a diverse mix of tree species, with 38.8% being coniferous, 33.4% deciduous, and 27.8% mixed forests (Korea Forest Service 2024). Fire data were collected from the National Statistical Yearbook of Forest Fires for 1991–2020 (Korea Forest Service 1992–2021). The date of occurrence, burned area, fire cause, and ignition location for each fire were retrieved from the annual forest fire report. The Korea Forest Service (KFS) has collected forest fire incident information across the country since 1991 and publishes an annual nationwide forest fire report. The locations of the fire incidents used in this study required geo-referencing before conducting the spatial analysis. ArcGIS Pro v.3.1 software (ESRI, Redlands, CA, <https://www.arcgis.com/index.html>) was used for coordinate transformation from street addresses to geographic coordinates. The location of fire ignition for 1991–2020 in South Korea is shown in Fig. 1.

Fires are often classified according to the burned forest area; however, there is no consensus on the classification criteria because of the wide variations in fire intensity, size, and type between countries and regions (Covington and Moore 1994). Different classification systems categorize fire size into different classes, with as few as two or up to six or more (Heinselman 1981; Ganteaume and Guerra 2018). In South Korea, until 2009, fire size was divided into three categories: small (< 5 ha), medium (5–30 ha), and large (> 30 ha). As fires have become larger and more frequent since 2010, fire control and suppression capacities have been reinforced by the use of firefighting helicopters. Simultaneously, the definition of a large forest fire was revised to include fires that can burn more than 100 ha of forestland. The small and medium fire classes are no longer used in national forest fire reports. Consequently, in this study, fires were grouped into small (< 5 ha), medium (5–100 ha), and large (> 100 ha) fires for analysis (Ganteaume and Guerra 2018), depending on the area burned by the individual fires.

To determine how the degree of climatic variability influences the risk of forest fire, common meteorological parameters (air temperature, precipitation, relative humidity, and wind speed) were used to examine the influence of climate on forest fire activity. Meteorological data were compiled from the data center of the Korea Meteorological Administration (<http://data.kma.go.kr>, accessed on September 8, 2023). The Pearson correlation coefficient was used to quantify the linear association between fire occurrence and climatic variables. The statistical analysis was conducted using the R programming platform (R Core Team 2023).

2.2 Temporal trend analysis

Trend analysis captures the general characteristics of time-series data. Previous studies have used several parametric and non-parametric approaches for trend analysis. Parametric approaches are more reliable than nonparametric ones, but they require data to be independent and normally distributed (Jiqin et al. 2023). As the number of fires and associated burned areas does not follow a normal distribution (Chatfield 1995), a nonparametric Mann–Kendall test is mostly used to detect monotonic upward or downward trends in fire statistics (Esterby 1996; Kendall 1975) due to its insensitivity to the normal distribution of data time series and outliers (Gilbert 1987). The temporal trend of fire statistics over time was analyzed using the Kendall rank correlation coefficient (τ) and its association significance level (p-value). The τ value can be used to determine the sign of the trend, i.e. upward ($\tau > 0$) or downward ($\tau < 0$). A trend was considered significant when the p-value was < 0.05 .



Fig. 1 Ignition points of forest fires for 1991–2020 in South Korea (BS: Busan, CB: Chungcheongbuk-do, CN: Chungcheongnam-do, DG: Daegu, GB: Gyeongsangbuk-do, GG: Gyeonggi-do, GN: Gyeongsangnam-do, GW: Gangwon State, JB: Jeollabuk-do, JJ: Jeju, JN: Jeollanam-do, SE: Seoul)

The magnitude of the change was subsequently assessed using Sen's slope estimator (Sen 1968), which determines the mean of all the pairwise slopes for any pair of points in the dataset. Sen's slope is considered better for detecting the statistical linear relationships in long-term temporal data (Jiqin et al. 2023). A positive value of Sen's slope indicates an upward trend, whereas a negative value indicates a downward trend.

The Mann–Kendall test and Sen's slope estimate were employed to examine the temporal dynamics of fire activity over a 30-year period (1991–2020). In addition, the temporal variation of fire occurrence in South Korea was analyzed over two time periods (1991–2005 and 2006–2020). The temporal trend of forest fires in South Korea was analyzed using the R programming platform (R Core Team 2023).

2.3 Fire season estimation

The fire season is defined as the time window in a year when fires are most likely to ignite and spread during the reference period (Chen et al. 2011; Pimont et al. 2023). However, there is no consensus on criteria for the start or end of the fire season (Wotton and Flannigan 1993). In previous studies, the exceedance date approach has been widely used to define the fire season from fire records. Specifically, the fire season is based on Julian dates each year when the weather variables (Jolly et al. 2015; Chen et al. 2011) or fire regime parameters (Silva et al. 2023; Pimont et al. 2023) cross predefined threshold values.

In this study, the exceedance date approach proposed by Silva et al. (2023) was used to identify the fire season based on the date of fire occurrence in a certain period. The days of the year that correspond to 5% and 95% fractions of significant fires (≥ 5 ha) in the cumulative sum of fires represented the start and end dates of the fire season (Hanes et al. 2019). Further, the difference in days between the start and end dates was regarded as the duration of the fire season. This provided a normalized indication of the annual distribution of fire events, which was independent of the outliers in the area burned each year. In this study, the total number of fire events could be counted from June to May of the successive year (12 months) because forest fires in the summer season (June–August) are extremely rare in South Korea. Additionally, forest fire prevention and management activities are terminated by the end of May, when the rainy season usually begins.

2.4 Spatial variation analysis

The spatial trends of fire statistics were examined separately for the first (1991–2005) and second (2006–2020) sub-periods. The spatial variability of fire occurrences was analyzed using hotspot analysis, and the spatial distribution pattern of large fires was detected using a standard deviation ellipse. Both KDE and Getis-Ord G_i^* hotspot analysis were used in this study to identify high-risk fire zones as areas with high concentrations of events. ArcGIS Pro v. 3.1 was used for spatial analysis.

KDE is a non-parametric method that has recently been used in many studies recently. In forest fire studies, KDE is used to create continuous fire occurrence density surfaces from the location of ignition points to represent the fire risk associated with the influencing factors. The kernel density function was defined as follows (Silverman 1986)

$$\hat{f}(x) = \frac{1}{nh^2} \sum_{i=1}^n K \left\{ \frac{s - s_i}{h} \right\},$$

where n is the total number of fire events, K is the kernel function, h is the bandwidth that defines the search radius of the kernel function, s is the location where the estimated density values are being calculated, and s_i is the location of each fire event.

The Getis-Ord G_i^* statistic is expressed by

$$G_i(d) = \frac{\sum_{j=1}^n w_{ij}(d) x_j - x \sum_{j=1}^n w_{ij}(d)}{s \sqrt{\frac{n \sum_{j=1}^n w_{ij}^2 - \left(\sum_{j=1}^n w_{ij} \right)^2}{n-1}}},$$

where x_j indicates the attribute value at location j , $w_{ij}(d)$ is a spatial weight matrix for all locations j within a certain distance d from the feature at location i , n is the total number of locations, $\bar{x}$ is the sample mean, and s is the sample variance. Inverse distance-weighted interpolation was used to generate a continuous-density surface from discrete Gi* estimates. This technique is widely used to map the spatial extent of hotspots.

Fire-dominant (hot spot) and fire-scarce (cold spot) areas were delineated using Z-scores (Feltman et al. 2012). For the Gi* statistic, the resultant Z-scores represent the degree of spatial dependence between the incident point data. The larger the Z-scores, the more intense the clustering of high values (hot spots), whereas more negative Z-scores indicate more intense clustering of low values (cold spots). A Z-score near zero indicates no apparent spatial clustering with a random distribution.

The standard deviation ellipse is commonly used to study the spatial distribution characteristics of geographical elements, display their spatial distribution patterns, and identify variations in the center of the elements (Lefever 1926). The concentration of fire events was assessed based on the elliptical area. The long semi-axis of the ellipse represents the main direction of the elemental distribution, whereas the size of the short semi-axis indicates the degree of spatial aggregation. The larger the difference between the lengths of the long and short axes, the more pronounced the directional distribution of the element. The azimuth represents the angle in a clockwise direction from the north to the direction of the long axis of the ellipse, indicating the element's main direction of distribution (Zhao et al. 2022). In this study, a standard deviation ellipse was used to quantitatively describe the spatial distribution patterns of large fires (> 100 ha) in South Korea.

3 Results

3.1 Long-term trends in forest fires

The number and burned area of fires in South Korea were aggregated annually. Figure 2 shows the uneven spatial distribution of fires and burned area from 1991 to 2020, which were largely derived from the annual cycle of meteorological parameters and human activities. For the 30 years from 1991 to 2020, 13,535 forest fires occurred in South Korea, with an average of 451 fire events per year. The highest number of forest fires was observed in 2001 (785 events), and the smallest number was recorded in 1991 (139 events). Approximately 2,085 ha of forests burned annually. Several large fires occurred in 2000–2001, followed by a sharp decreasing trend to 2013. An extreme example was in 2000, when the East Coast Fires of 2000 were responsible for 23,794 ha of burned forest lands that exceeded previous records. Over the 30 years, approximately 95% of the recorded fires in the national dataset were small (< 5 ha); however, these accounted for only approximately 20.4% of the total burned area. In contrast, large fires (> 100 ha) represented 72.4% of the total area burned, which constituted only 0.4% of all fires. The disproportionately significant contribution of large fires to the total burned area has been commonly observed in other countries (Díaz-Delgado et al. 2004; Scott et al. 2017; Bountzouklis et al. 2022).

The Mann–Kendall test was conducted to detect annual trends for both the number of fires and the area burned over the 30 years (Table 1). The annual fire frequency followed an upward trend (Kendall $\tau=0.177$) with spikes in two consecutive years, 2000–2001;

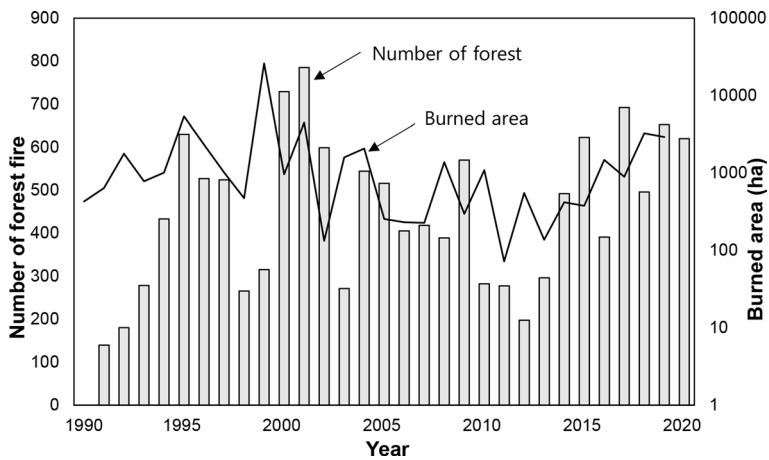


Fig. 2 Temporal variations and trends in the number of fires and burned area from 1991 to 2020

Table 1 Forest fire statistics of South Korea for 1991–2020

Fire regime	Trend	All fire	Small fire (< 5 ha)	Medium fire (5–100 ha)	Large fire (> 100 ha)
Fire occurrence (event/yr)	Mean	451.2	428.6	20.8	1.8
	Kendall τ	0.177 ($p=0.175$)	0.228 ($p=0.080$)	-0.440 ($p=0.001$)	-0.051 ($p=0.733$)
	Sen's slope	5.82	7.26	-1.00	0
Area burned (ha/yr)	Mean	2085.8	243.8	330.0	1511.3
	Kendall τ	-0.08 ($p=0.544$)	-0.52 ($p=0.000$)	-0.31 ($p=0.017$)	-0.03 ($p=0.637$)
	Sen's slope	-12.08	-9.66	-10.21	-4.45

however, this pattern was not significant ($p=0.175$). Table 1 shows an increasing trend for small fire occurrence ($\tau=0.228$, $p=0.080$), but decreasing trend for the annual frequency of medium fires ($\tau=-0.440$, $p=0.001$). The increasing trend in the annual total fire occurrence is associated with increased surface fuel accumulation and climate change; however, the negative trend in the number of medium and large fires was due to the massive early fire suppression drive in South Korea (Han 2021).

There was an annual linear decrease in the forest area burned in South Korea over the past 30 years, with significant declining trends in small ($\tau=-0.52$, $p<0.001$) and medium ($\tau=-0.31$, $p=0.017$) fires. According to Sen's slope estimator, areas burned by small and medium fires decreased by 9.66 and 10.21 ha annually, respectively. The number and burned area by large fires did not show any significant trend for the entire period, because the frequency of large fires was too small for statistical analysis.

The number of fire events showed a significant increasing trend in the 15-year sub-periods, with Kendall $\tau=0.333$ ($p=0.092$) for 1991 to 2005 and $\tau=0.35$ ($p=0.075$) for 2006 to 2020. The average annual burned areas were 48,970.98 ha for 1991–2005 and 13,582.72 ha for 2006–2020. Slight upward trends in burned area were observed in the first ($\tau=0.200$, $p=0.322$) and second ($\tau=0.3410$, $p=0.038$) sub-periods. From 1991 to 2005, Sen's slopes

were 24.18 events/yr for fire occurrence and 78.99 ha/yr for burned area, while these values were 18.16 events/yr and 74.65 ha/yr, respectively, for the second sub-period.

The monthly variations in fire occurrence and burned area are shown in Fig. 3. The warming stripes diagrams in Fig. 3 were used to depict the temporal evolution of fire frequency and burned area over time. Fire occurrence was concentrated in April, with 130 fires per year, and least common in July, with only 2.3 events per year. A total of 61.83% of fires occurred from March to May (spring), while summers (June to August) were less active, with only 5.52% of fires.

The months of high fire activity followed a unimodal distribution, with a peak in April, mainly representing the dry season in a monsoonal climate. The number of large fires (> 100 ha) peaked in April (36 events), followed by March (13 events) and May (3 events). The monthly distribution of burned area over the entire study period showed broad concentrations in April and May, when new leaves start to emerge in the spring. These two spring months accounted for 80.70% of the total area burned annually, mostly due to the warmer temperature, low humidity, and extreme dryness (Jeon and Chae 2017).

Distinct seasonal patterns were observed, with higher occurrences in the drier part of the year, spring and winter, and limited fire occurrence in the summer (Fig. 4). The incidence of fire in the spring was significantly higher than in winter, with the highest number of forest fires occurring in 2000. In the spring, when precipitation is low and leaf unfolding has not yet started, the fuel moisture deficit and dry weather lead to high-intensity fires, particularly in coniferous forests (Baek et al. 2022). Moreover, the warmer air temperature in spring could increase human presence and activity in forests, which could enhance the risk of accidental and intentional ignition (Canadell et al. 2021). In contrast, fires that occur in the summer are usually smaller and fewer due to the higher number of rainy days.

As shown in Table 2, an upward trend in fire frequency was detected in all seasons. Significant upward trends in the number of fires were detected in the fall (Kendall's $\tau=0.267$, Sen's slope=0.67) and summer (Kendall's $\tau=0.513$, Sen's slope=1.44). A wide variation in the burned area existed in the four seasons, but no statistically significant trend was observed throughout the study period. Nevertheless, the areas burned in spring, fall, and winter showed slight downward trends from 1991 to 2020. Although the frequency and magnitude of fires in the summer exhibited increasing trends, the influence on the forest fire regime was limited.

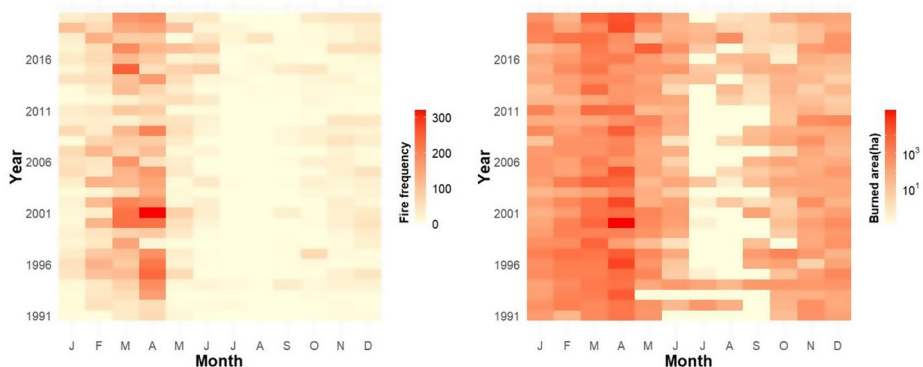


Fig. 3 Monthly variation in fire regime for the study period of 1991–2020

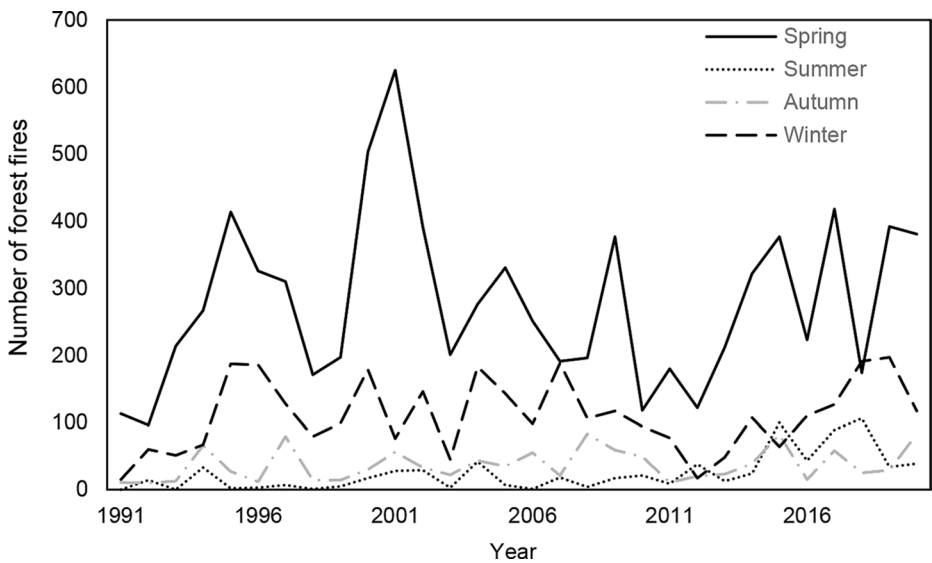


Fig. 4 Seasonal variations in fire occurrences from 1991 to 2020

Table 2 Mann–Kendall test and Sen’s slope estimator results

Fire regime	Trend Analysis	Spring (March to May)	Summer (June to August)	Fall (September to November)	Winter (December to February)
Fire occurrence (event/y)	Mean	278.96	24.9	37.23	110.07
	Kendall’s τ	0.101	0.513**	0.267*	0.156
	Sen’s slope	2.00	1.44	0.67	1.36
Area burned (ha/y)	Mean	1927.43	10.94	23.92	132.51
	Kendall’s τ	−0.071	0.177	−0.361	−0.241
	Sen’s slope	−10.82	0.15	−0.60	−3.80

* p value < 0.05, ** p value < 0.01

Table 3 Exceedance dates of fire season estimations based on the number of fires

Period	Exceedance date of fire occurrence		Length of fire season
	5%	95%	
1991–2020	December 3	April 28	147
1991–2005	December 5	April 28	145
2006–2020	November 19	May 6	169

3.2 Fire concentration

The fire season is generally defined as the time of year that accounts for most of the fire activity. Over the 30 years, 5% of forest fires (> 5 ha) occurred before June 11, and another 5% happened after October 3 (Table 3). The length of the forest fire season became elongated from 144 days in the first sub-period (1991–2005) to 169 days in the second sub-period

(2006–2020). The forest fire season of the second sub-period started 17 days earlier and ended 8 days later than that in the first sub-period. The increased temperature and reduced precipitation contributed to longer and more intense winter and spring droughts (Jolly et al. 2015), which resulted in the lengthening of the fire season. Over the past 30 years, the peak season of fire incidents has shifted approximately 10 days earlier, specifically from early April to mid-March. The rapid warming and little snowfall due to climate change had led to earlier mountain snowmelt in spring, and consequently caused an earlier peak of forest fires (Westerling et al. 2006).

An increase in fire season length will potentially lead to a higher number of fires and has stimulated greater demand for fire prevention and suppression activities over a much longer period in recent years. This raises the issue of fatigue in fire watching and suppression resources (Pimont et al. 2023; Wotton and Flannigan 1993).

3.3 Fire–climate relationships

The relationships between meteorological variables and fire frequency during the forest fire season (January to May, November to December) of South Korea were examined (Fig. 5). Burned area is particularly important, but is mostly influenced by the management and suppression efforts. Therefore, only the influence of climate variability on fire ignition density was analyzed. The spatial variation in climate variables over time was not considered in the analysis.

Figure 5 shows that, during the forest fire season, both the precipitation amount ($R=0.628$, $p<0.001$) and number of rainy days ($R=0.669$, $p<0.001$) were significantly related to forest fire activity. This indicates that precipitation deficits seem to be the principal determinant affecting the occurrence and distribution of forest fires. Air temperature and relative humidity are known to influence fuel moisture, but weak correlations with fire frequency were observed (Fig. 5(a) & (f)). Wind speed had no significant effect on fire occurrence (Fig. 5(e)). It is generally agreed that wind speed has a stronger influence on the behavior of a fire than the number of fire events (Benson et al. 2008; Carmo et al. 2022). Regardless of statistical significance, the temperature and number of dry days were positively related to fire occurrence, whilst other meteorological variables, such as precipitation, humidity, and wind, were negatively related to the number of fires. However, there is considerable variability in meteorological variables with fire activity. This is consistent with previous studies that reported an increased frequency of occurrence in recent decades due to reduced precipitation and higher temperatures caused by climate change (Canadell et al. 2021; Dastour et al. 2024; Jeon and Chae 2017).

Figure 6. Summarizes the anomaly analysis conducted on annual fire and climate dataset, which represents an anomalous bases on the standard anomaly throughout the study period. The area burned is not directly related to the number of forest fires in the 1991–2005 period, but there is a good correlation in the 2006–2020 period. Figure 6 exhibits that during the first half period (1991–2005), precipitation related weather had distinctive contribution on fire activity, but its influence may not be valid for the second half period. In contrast, air temperature anomalies are correlated with the temporal variations of fire activity for 2006–2020.

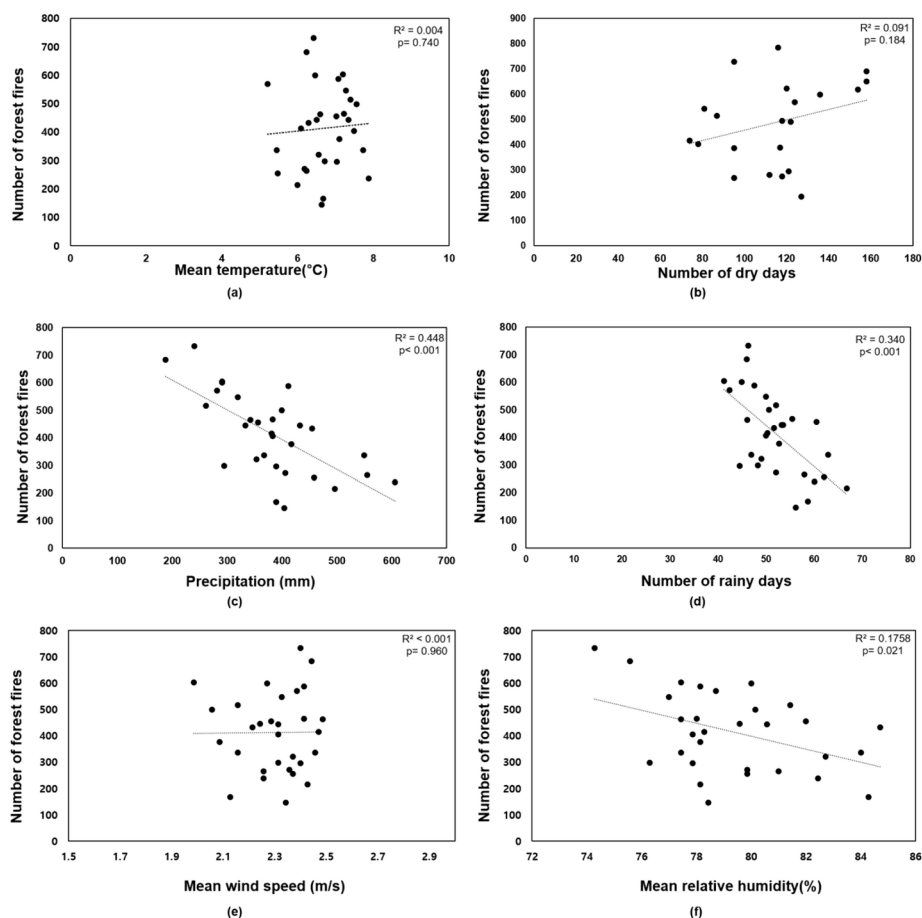


Fig. 5 Effects of climate variables on fire occurrence during the fire season; (a) mean temperature, (b) number of dry days, (c) precipitation amount, (d) number of rainy days, (e) mean wind speed, and (f) mean relative humidity

3.4 Spatial clustering analysis of forest fires

The spatial hotspots of fire ignition across the country were analyzed for two 15-year periods (1991–2005 and 2006–2020). The term hotspot or hot region refers to a geographic area that has had a greater than average number (or area) of forest fires, or an area where the average risk of large fires is higher. In this study, 10×10 -km forest grids were generated for the hotspot analysis. Small fires (<5 ha) were excluded from the spatial analysis because they did not result in large cumulative burned areas. The spatial analysis of forest fire characteristics over each 15-year span did not consider the time effect. Ignition locations are not necessarily uniform across the country, but may be clustered in certain regions, mainly due to ignition sources, fuel type and load, climate, and other factors.

The continuous surface of the KDE maps was used to visualize the spatial distribution pattern of the forest fire events for different time periods. Figure 7 shows that different clus-

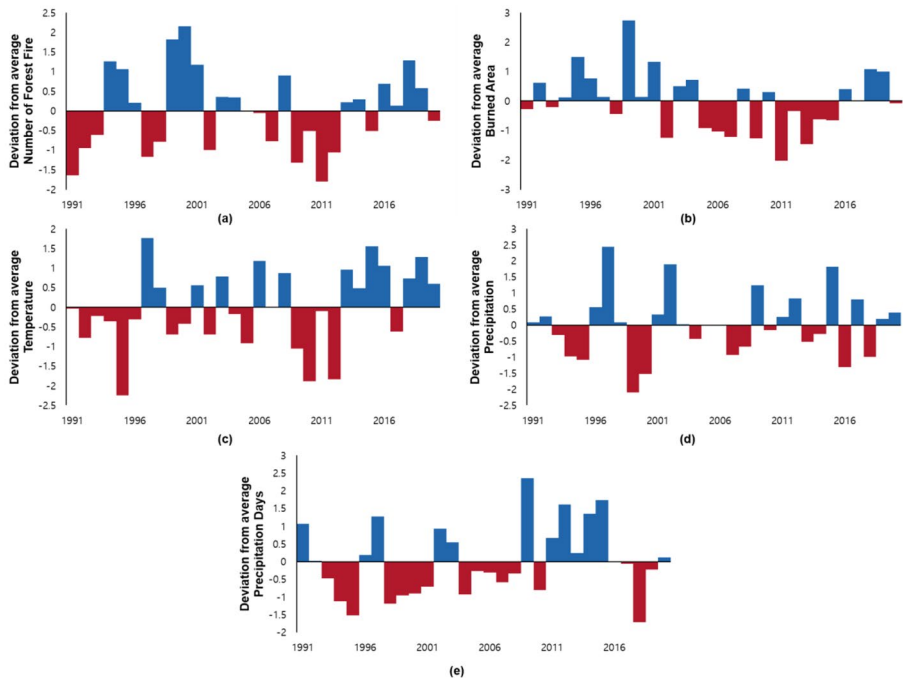


Fig. 6 The relationship between climate anomalies and forest fire activity; **(a)** deviation from average number of forest fire, **(b)** deviation from average burned area, **(c)** deviation from average temperature, **(d)** deviation from average precipitation, **(e)** deviation from average precipitation days

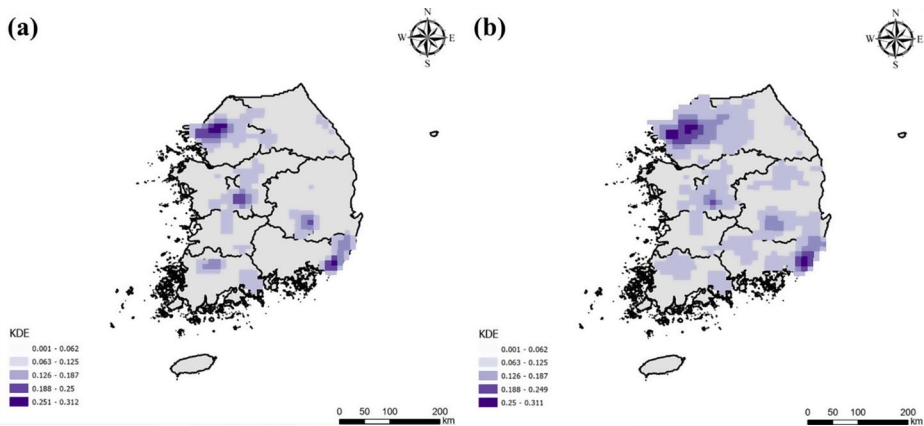


Fig. 7 KDE maps of forest fires **(a)** 1991–2005, and **(b)** 2006–2020

tering patterns of fire locations were observed both spatially and temporally. As expected, most of the forest fires occurred near the cities of Seoul and Busan. The number of fire activity near Seoul and Busan accounted for 282(4.19%) and 288 (4.28%) ignition points in the 1991–2005 period, respectively, and the values increased to 157 (2.31%) and 242 (3.56%)

locations in the 2006–2020 period. This can be attributed to the increased outdoor and recreational activities near to residential areas.

Figure 8 also shows the Getis-Ord G_i^* statistics of fire occurrence for the two sub-periods. The spatially interpolated surface was classified into seven different hotspot classes based on natural break techniques in ArcGIS Pro V.3.1. High spatial concentration was apparent around the largest cities, such as Seoul, Daejeon, Daegu, and Busan. They are highly populated and the number of recreational activities in forests is increasing, especially in spring; in these areas, fire occurrence was highest. This could be because all forest fires in South Korea are triggered by human ignition, such as arson, negligently discarded cigarettes, and the burning of agricultural debris (Korea Forest Service 2022).

There was a slight difference between the two periods, with the second one having significantly distinct hotspots in the Seoul metropolitan and Busan regions, which are more spatially dispersed. Notably, a considerably larger part of these regions was exposed to forest fires during the study period. The hotspots analysis for 2005–2020 showed that the western and southern regions along the coast could be considered cold spots because most of the land is covered with rice paddy that is cultivated in the rainy season; therefore, the surface fuel loads to burn are low in the winter and spring (Lamat et al. 2021).

Areas that are characterized as high, moderate, and low require different levels of attention. In Fig. 8, the very high (red) areas indicate those that require more attention from the government for the management of forest fire incidence, as they are more congregated with high Z-scores. In contrast, very low areas (blue) represent a significant clustering pattern of negative Z-scores. These areas require the least attention because they are considered cold spots.

The location of fire incidents is an important factor to consider for planning preparedness, prevention, and pre-suppression (initial attack) activities. However, when focusing on large fires that burn most of the land area, the start locations, although still clustered, may show a different spatial pattern to all the fires. The behavior of large fires depends on the fuel conditions (some fuel types are more prone to rapid spread), remoteness from suppression resources, and other factors, such as resistance to control.

To examine the spatial dispersion of large fires (> 100 ha) over 15 years, a standard deviation ellipse analysis was conducted. The area of the standard deviation ellipse varied over time and was, in general, greater for the first sub-period than for the second sub-period. As shown

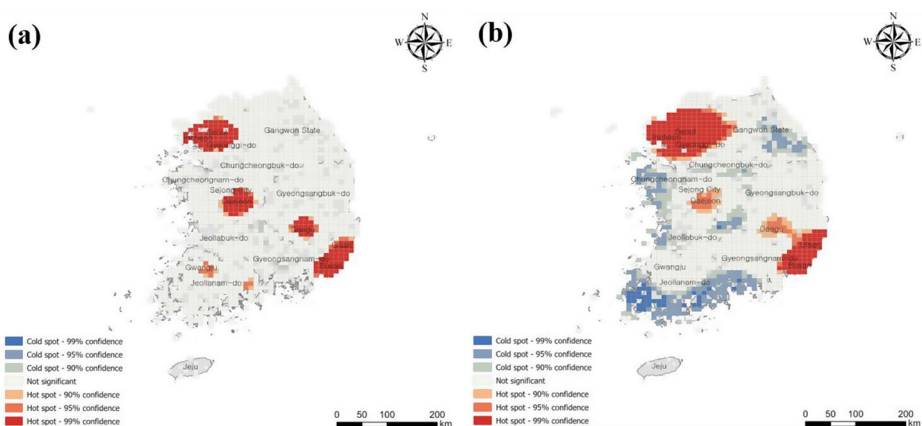


Fig. 8 Hotspots map of forest fire occurrence for (a) 1991–2005 and (b) 2006–2020

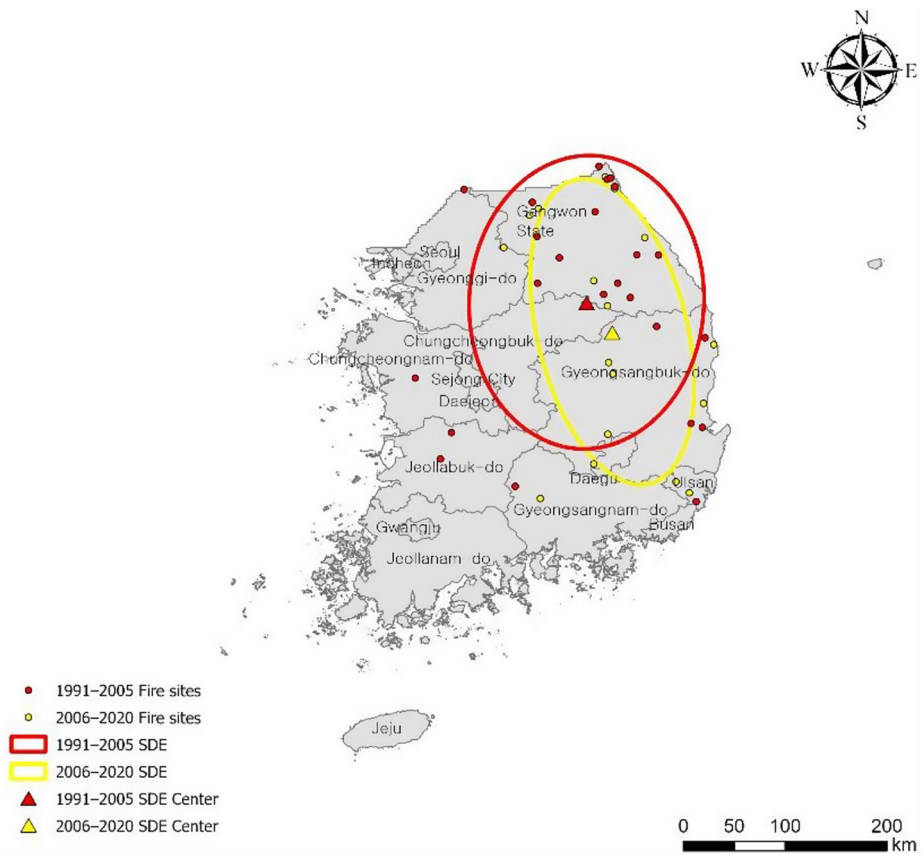


Fig. 9 Standard deviation ellipses of large fires for 1991–2005 and 2006–2020

in Fig. 9, significant changes were observed in the lengths of the major and minor axes of the ellipses between the 1991–2005 and 2006–2020 sub-periods. The spatial dispersion of the 24 large fires in the first sub-period was north–southward over an area of 42,200 km². During the second period, the major axis of the ellipse lengthened by 16 km, whereas the minor axis shortened by 76 km. The spatial changes in the minor and major axes indicate a north–south directional expansion of forest fire occurrences, whereas the east–west dispersion was less wide. In the second sub-period, large fire activity extended over portions of Gangwon and Gyeongsangbuk Provinces.

Large fire occurrences exhibit a strong directional pattern during different periods. From the first sub-period to the second, the center of the ellipse shifted 35.33 km in the southwest direction with a deviation angle of 16.95°, indicating that large fire activity tends to mostly occur along a northwest–southeast major axis. According to the ellipse of large fire occurrences, the risk of large fires is increasing and becoming more concentrated in Gangwon and Gyeongsangbuk Provinces, accounting for 41.1% and 32.8% of the total burned area during the second 15-year period.

The northeastern region mostly lies in mountainous forests with a higher growing stock (186.9 m³/ha in 2021), and they frequently experience strong winds (>25 m/s wind speed). The

mountainous physiography of the northeastern region makes it more difficult to control and regulate fire events. The larger fuel load and extreme weather increased the risk of large fires. In addition, a good deposition of inflammable dry pine leaves acts as a fuel for the rapid spread of fire lines. The dominant tree species in the northeastern region is red pine (*Pinus densiflora*), whose needles are highly flammable due to their resin, which accelerates the ignition and fire intensity (Li et al. 2021; Baek et al. 2022). This could result in a higher incidence of uncontrolled fires and potentially larger burned areas.

3.5 Effect of human activity on Forest Fire trends

Human activities are a major cause of forest fires in South Korea. Approximately half of all forest fires were caused by human negligence during outdoor activities and by the burning of discarded cigarettes (Korea Forest Service 2022). Holidays have been extended since 2004 with the introduction of the 5-day work week system in South Korea. This allows for long weekends, which in turn encourages outdoor activities in the forest. The proportion of fire frequency on weekends increased from 35.0% in 1991–2005 to 36.6% in 2006–2020. This is mainly due to increased outdoor activities combined with the 5-day working week and warmer and drier weather in spring.

Another major cause of forest fires was Arbor Day. In South Korea, 5 April has been designated as Arbor Day, a national holiday since 1949, to encourage people to plant trees. On the 5th of April each year, large numbers of people went to the forest to take part in tree planting activity, which unfortunately led to human-caused fires. Arbor Day was excluded as a public holiday in 2006. The total number of forest fires that occurred on 5 April was 331 events in 1991–2005 and 78 events in 2006–2020, a reduction of 76.4%.

4 Discussion

South Korea has experienced frequent forest fires in recent decades. The combined effects of climate, fuel, and human factors, which may influence the initial ignition, fire intensity, and duration of the fire season, contributed to the increasing number of forest fires in South Korea (Jo et al. 2023).

Climate change has led to an increase in fire activity by altering fire weather conditions (Jolly et al. 2015). Most previous studies have found a significant increase in fire frequency and intensity worldwide due to climate change (Flannigan et al. 2006, 2013; Kelly et al. 2013). They investigated how climate variables such as temperature, precipitation, and wind speed affect the occurrence and intensity of forest fires (Dastour et al. 2024). As in other countries, reduced precipitation and resulting water deficits could increase the dryness of combustible fuels, making forests more vulnerable to fire. During the study period, both the amount of precipitation and the number of rainy days had a significant relationship with fire frequency. This is in line with findings from previous studies in other countries (Canadell et al. 2021; Dastour et al. 2024). Warmer temperature is a key variable influencing increased fire risk, but was not statistically validated in this study ($p=0.740$). This is mainly due to the relatively high variability of the temperature, with a narrow range of fluctuation (5–8°C, Fig. 5). The longer duration of fire season also contributes to the high ignition risk in forests. This phenomenon is predicted to spread across the country,

with longer, drier springs, and shorter, wetter summers. This changing climate, driven by global warming, is expected to affect the occurrence, intensity, and seasonality of fires in South Korea.

Fuel load, which refers to the amount of combustible material in forests, is a key variable influencing the likelihood of fire. Since 1973, the national reforestation project has led to successful restoration, resulting in an accumulation of fuels (Korea Forest Service 2024). South Korea's forests are currently characterized by dense vegetation due to insufficient thinning and clearing. This dense forest, with thick accumulations of leaf litter on the forest floor, creates a highly combustible environment in which fires can easily ignite and spread rapidly (Chang et al. 2024). The northeastern region of South Korea has faced more frequent and severe forest fires, especially those caused by the greater fuel load. Changes in local climate and weather patterns have direct or indirect effects on forest fuel loads by altering tree species distribution, growth rates and overall forest structure (Dastour et al. 2024). However, a complex relationship between forest ecology and climate change is not explored in this study.

There is evidence that climate has a greater influence on fire risk than human activities in European, Oceanian, and American countries (Hanes et al. 2019; Miller and Safford 2012; Díaz-Delgado et al. 2004). On the contrary, forest fires in South Korea are primarily caused by anthropogenic factors. Human mistakes were the most frequent cause of fires (49%), followed by negligence during agricultural work (18%) and cigarette smoking (10%) (Korea Forest Service 2022). High air temperatures and low rainfall could increase human presence and activity in forests, leading to a high risk of accidental and intentional ignition (Curt et al. 2016). The frequent fire incidents have occurred in suburban forests near cities because of the increasing demand for forest recreation. This proximity to human settlements has enabled earlier detection and faster suppression responses to fire ignition.

Forest fires have not been burning more intensely nor charring more areas in recent decades. This study showed that the destruction caused by forest fires slightly increased in the first half of the study period (1991–2005), but decreased in the second half. When ranking the years according to area burned, the top five worst years occurred in the first sub-period.

One possible explanation for this adverse trend is the forest fire surveillance and initial attack systems (Han 2021). The KFS is authorized to perform immediate suppression actions, and fire suppression policies have been enforced since the 2000s. The fire management policy was reinforced by fire prevention programs after the disastrous fires in 2000, when more than 20,000 ha were burned in the eastern coastal region.

The East Coastal Fires of 2000 scorched 23,794 ha of land and was the most destructive forest fire event ever recorded in the modern history of South Korea. After the 2000 forest fires, the KFS recognized that fire prevention strategies and practices, rather than fire suppression, would be more effective to mitigate the damage to people's property and lives. Therefore, the KFS developed a forest fire warning system that could inform of the real-time forest fire danger index based on vegetation, topography, and weather condition. Additionally, huge firefighting helicopters (S-64E) were first operated in 2002 to rapidly and efficiently extinguish a forest fire. Consequently, increased investment in fire suppression might have played a role in reducing the burned area in South Korea, similar to other countries (Stocks et al. 2002; Miller and Safford 2012).

Our findings have deepened our understanding of forest fire risk, but this study has certain limitations that should be addressed. First, this study analyzed the spatial and temporal trends in fire activity using historical fire records that are valid for the present, but potentially invalid for the future under a changing climate. The climate conditions of the future are highly uncertain and may differ from what has been observed in the past. This means that the fire trends that are

derived from the historical record are not projected into the future. Nevertheless, an understanding of how forest fires have changed over time, in terms of frequency, intensity, and geographical distribution, is critical to planning and implementing forest management strategies to reduce fire risk.

Climate warming is expected to increase the duration and severity of dry seasons, contributing to prolonged moisture deficits in fuels, which in turn will create a more fire-prone environment in South Korea. However, this study does not fully explain the mechanisms and processes underlying the relationship between fire activity and these climatic factors. To better understand the relationship between fire activity and climate variables, further research is needed to explore how changes in temperature, precipitation patterns, humidity, and vegetation moisture interact to influence fire behavior. Factors such as fuel moisture, fuel load, and fuel composition should also be studied to understand how these elements combine to create conditions that are conducive to forest fire.

Our results show clear evidence of an increased fire season in the future due to ongoing climate change, which is expected to continue over time. Although landscape-level fuel build-up cannot be addressed, this study can provide valuable insights for forest fire managers and agencies to properly allocate fire suppression strategies and resources. By understanding the projected changes in fire season length and severity, decision makers can better prepare for future fire risks and ensure timely and effective responses.

5 Conclusion

Fires are the most destructive agent in forest ecosystem worldwide. This study highlights the temporal and spatial trends in fire frequency and magnitude across South Korea for 1991–2020. A trend analysis was conducted using the Mann–Kendall and Sen's slope tests for different periods of time and fire size classes. The change in the spatial dispersion of fire activity was examined using hotspot analysis and the standard deviational ellipse.

The analysis of the 1991–2020 period revealed an increasing trend in the annual number of forest fires and the area burned in the country. Due to climate change, the temporal concentration of fire occurrences has shifted from May to April, and summer fires have gradually increased from 0.2% of total events in the 1990s to 10.0% in the 2020s. Hot and cold local administrative districts over the country were also classified based on hot spot analysis of the 30-year fire database. Large fire occurrences exhibited a strong directional pattern from the first sub-period (1991–2005) to the second (2006–2020), with the center of the ellipse shifting 35.33 km in the southwest direction with a deviation angle of 16.95° . This indicates that large fire activity tends to mostly occur along the northwest–southeast major axis, and the risk of large fires is increasing and becoming more concentrated in Gangwon and Gyeongsangbuk Provinces.

Owing to climate change and forest recovery, the number of fires increased throughout the period; however, no trends were observed in the burned area. Variations in fire statistics depend on many factors, such as weather, topography, land use, fuel characteristics, and human activity. Therefore, further studies on the relationships between these influencing factors are required to properly manage forest fire incidents occurring in South Korea. Additional research is also needed to better understand the increasing frequency of mega-fires and potential links between their increased frequency and changes to their spatial dispersion and distribution.

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Declarations

Conflict of interest The authors declare no conflicts of interest.

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